

Alpha Carbon Chemistry: Enols and Enolates

Organic Chemistry II

CHEM 2312

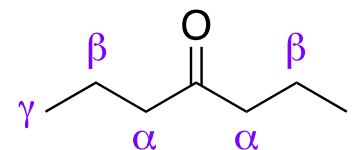
Dr. Michael Evans

1. Rationalize the acidity of alpha carbons of carbonyl compounds using resonance and inductive effects.
2. Draw enol tautomers of a given carbonyl-containing molecule and predict the most stable enol tautomer; describe the mechanism of tautomerization using elementary steps.
3. Recognize and apply the structural features that favor enolization.
4. Recognize and apply enolates as ambident nucleophiles.
5. Apply an appropriate base to generate an enolate or stabilized enolate with appropriate regiochemistry.
6. Predict the products and describe the mechanisms of alpha-halogenation reactions of ketohydes, including haloform reactions of methyl ketones.
7. Predict the products and describe the mechanisms of S_N2 alkylations of enolates with primary alkyl halides.
8. Apply the malonic ester synthesis for the preparation of substituted carboxylic acids.
9. Apply the acetoacetic ester synthesis for the preparation of substituted methyl ketones.

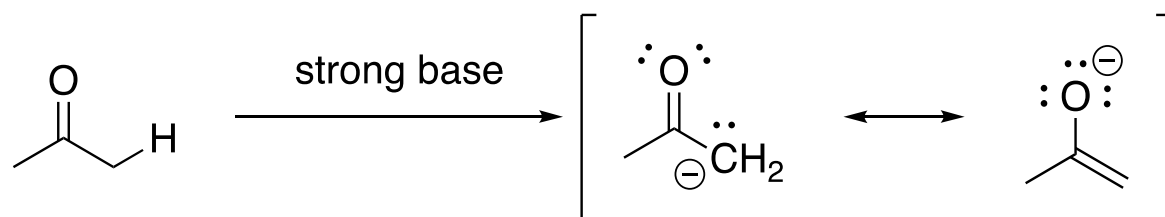
10. Predict the products and describe the mechanisms of aldol reactions of ketohydes.
11. Apply strategies for *selective* crossed aldol reaction of two different aldehydes or ketones, including Claisen-Schmidt condensation, directed aldol reactions, and intramolecular aldol reactions.
12. Predict the products and describe the mechanisms of Claisen condensations of esters and related acylations of enolates.
13. Apply strategies for *selective* crossed Claisen reaction of two different esters, including use of a non-enolizable ester and intramolecular (Dieckmann) condensation.
14. Predict the products of nucleophilic additions to α,β -unsaturated carbonyl compounds, which may undergo either direct (1,2-) or conjugate (1,4-) addition.
15. Predict the products and describe the mechanisms of Michael addition reactions.
16. Apply the Robinson annulation, a reaction that combines Michael addition and aldol condensation, in synthesis and retrosynthesis.

21.1 Acidity of the Alpha Carbon

- Carbons near a carbonyl group are designated using Greek letters



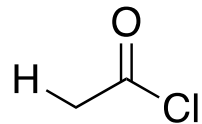
- Because the carbonyl group is electron withdrawing by resonance, the α positions are mildly acidic (pK_a 20...ish)
- The conjugate base of a carbonyl compound (an **enolate**) is resonance stabilized



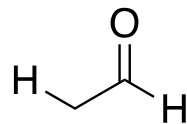
- Protonation of an enolate at oxygen yields a constitutional isomer of the starting carbonyl compound, an **enol**

Relative Acidity of Carbonyl Compounds

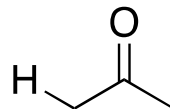
- Relative acidity of carbonyl compounds depends on the electrophilicity of the carbonyl carbon



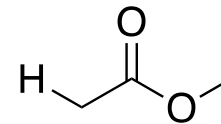
acetyl chloride



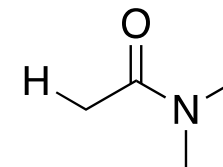
acetaldehyde



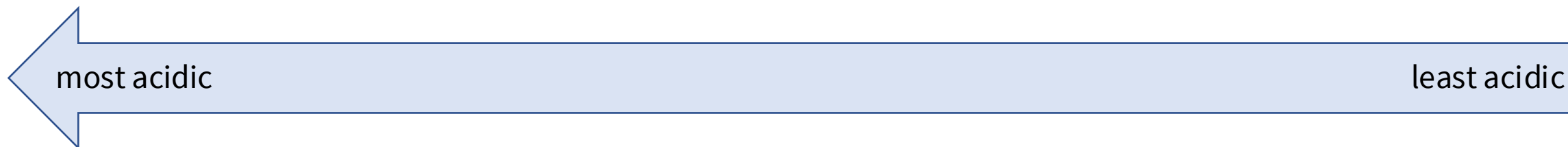
propan-2-one



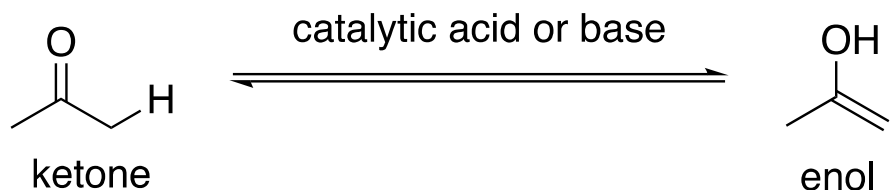
methyl acetate



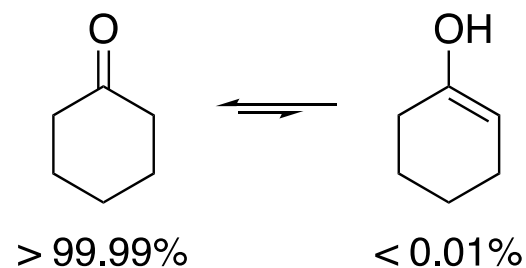
N,N-dimethylacetamide



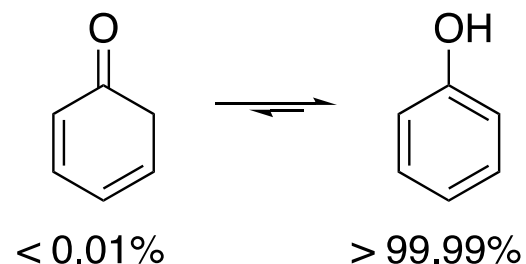
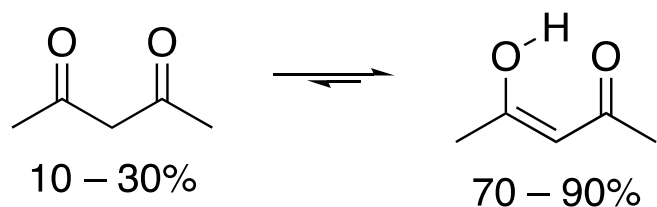
- In the presence of catalytic acid or base, enols are in equilibrium with the **keto** isomer
- Enol and keto isomers are known as **tautomers**, which differ in the position of a hydrogen in a conjugated system containing a heteroatom



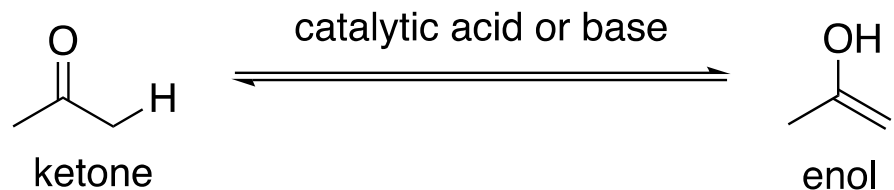
- The keto tautomer is heavily favored in most cases



- Conjugation, aromaticity, and hydrogen bonding can shift the position of equilibrium toward the enol

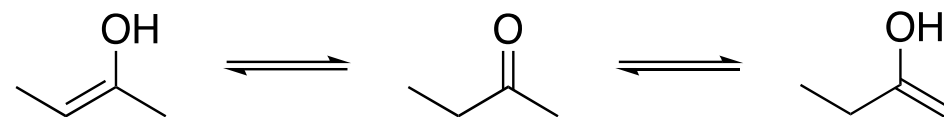


- The interconversion of tautomers is called **tautomerization**

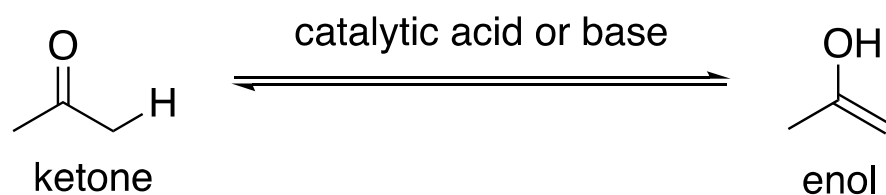


- The mechanism involves two sequential proton transfers, catalyzed either by acid...
- ...or by base:

- When multiple isomeric enols are possible, the more stable enol has the more substituted double bond

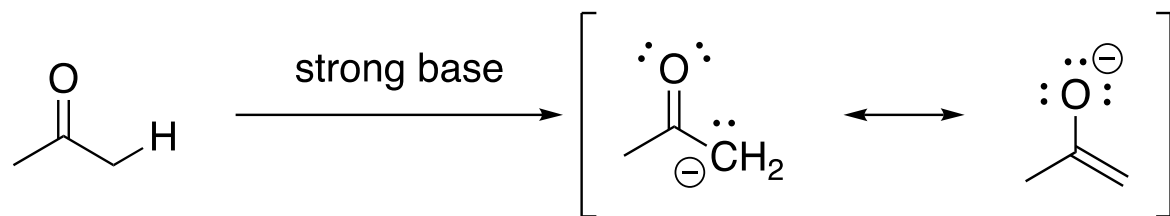


- Although the enol is often present in small amounts, it is very reactive as a nucleophile at the α position

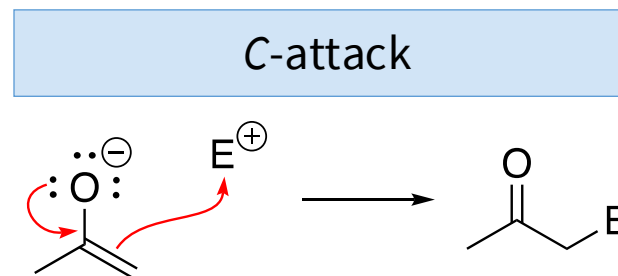
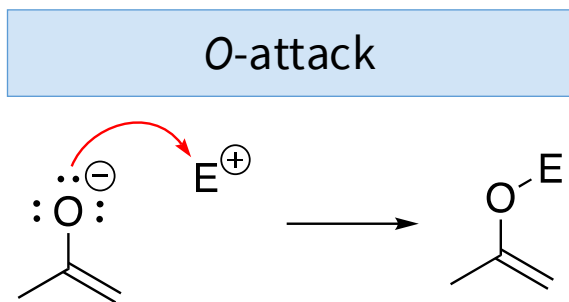


21.1 Generation and Reactions of Enolates

- Like enols, enolates are profoundly nucleophilic at the α position
- They are also nucleophilic at oxygen (**ambident** nucleophiles); consider the two major resonance contributors



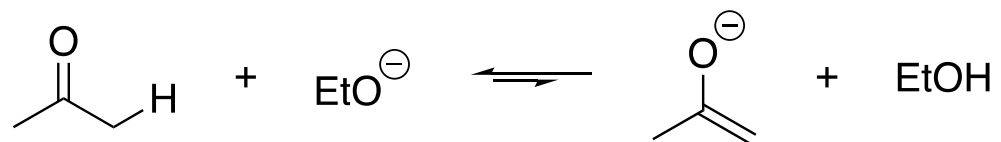
- Electrophiles may react with the carbonyl oxygen (O-attack) or the α carbon (C-attack); the latter is more common



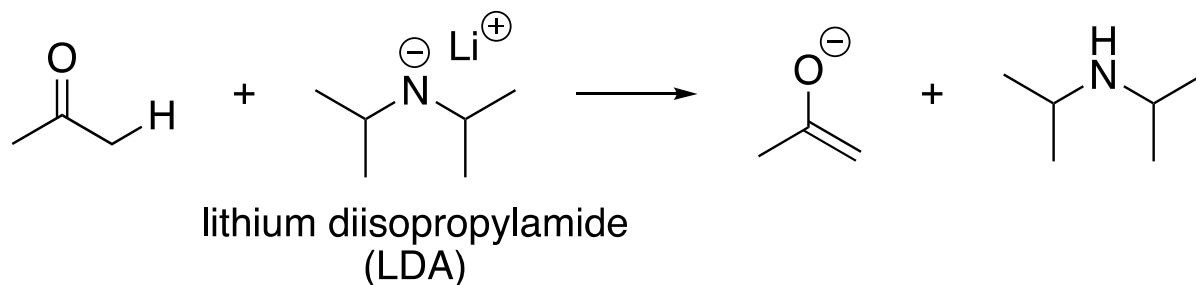
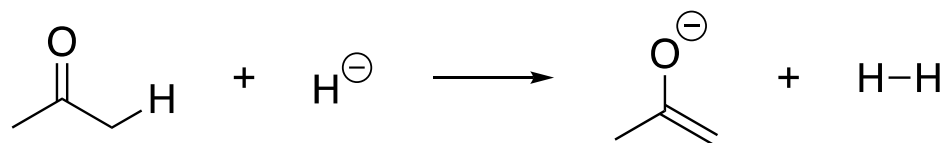
- Keep in mind that only the α position is acidic

21.1 Choosing a Base for Enolate Formation | 1 of 2

- Carbonyl compounds are not *that* acidic...alkoxide bases only deprotonate them to a small extent

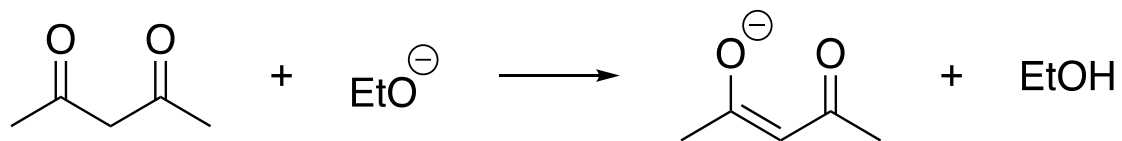


- Hydrogen and nitrogen anions are basic enough to completely deprotonate carbonyl compounds

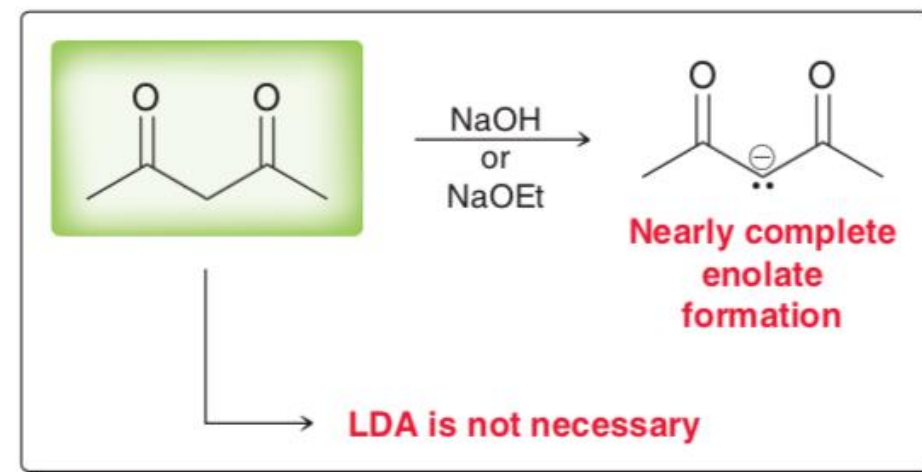
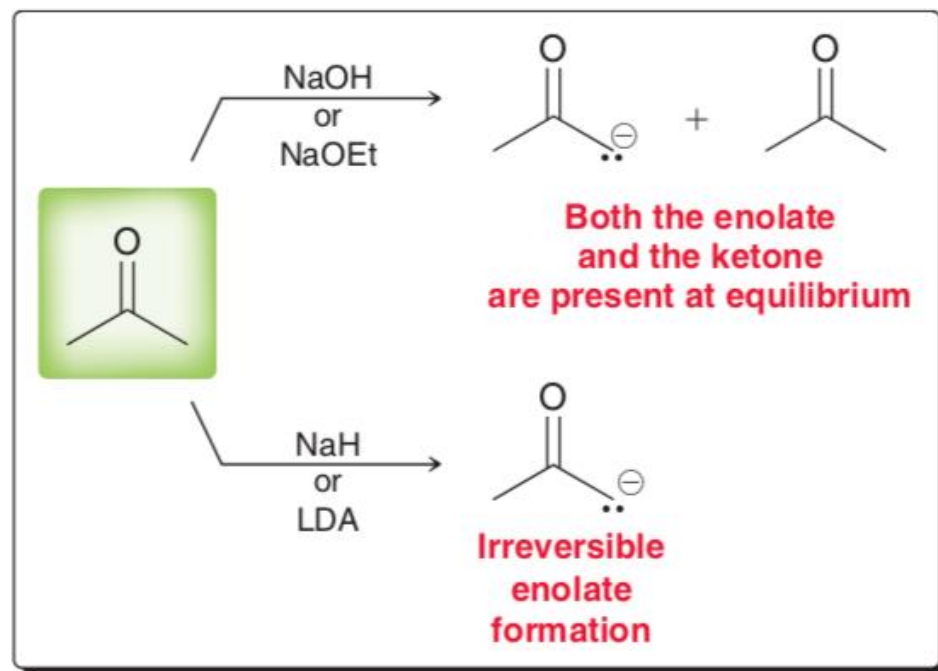


21.1 Choosing a Base for Enolate Formation | 2 of 2

- For 1,3-dicarbonyl compounds and other **activated methylene** compounds, alkoxide bases work very well!

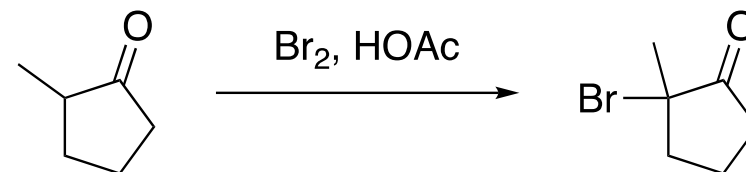
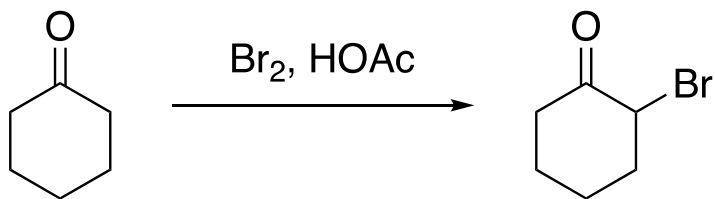


- In summary,



21.2 Alpha Halogenation of Enols

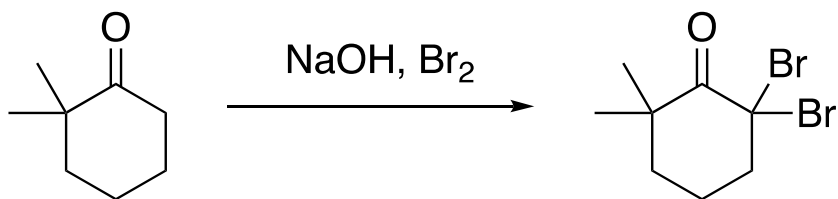
- Ketones and aldehydes undergo halogenation at the α position under acidic conditions via an enol intermediate
 - Enol formation is rate determining
 - When there is a regiochemical issue, the more substituted α -haloketone is the major product



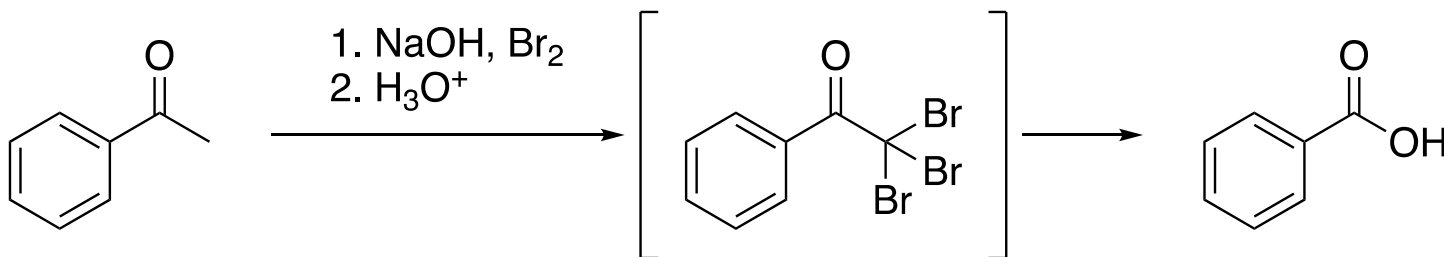
- The mechanism involves acid-catalyzed tautomerization (1. pt; 2. pt) followed by association of the electrophile and loss of a proton from the carbonyl oxygen

21.2 Alpha Halogenation of Enolates

- Under basic conditions, all α hydrogens are replaced with halogens
- This is not synthetically useful unless a methyl ketone is used

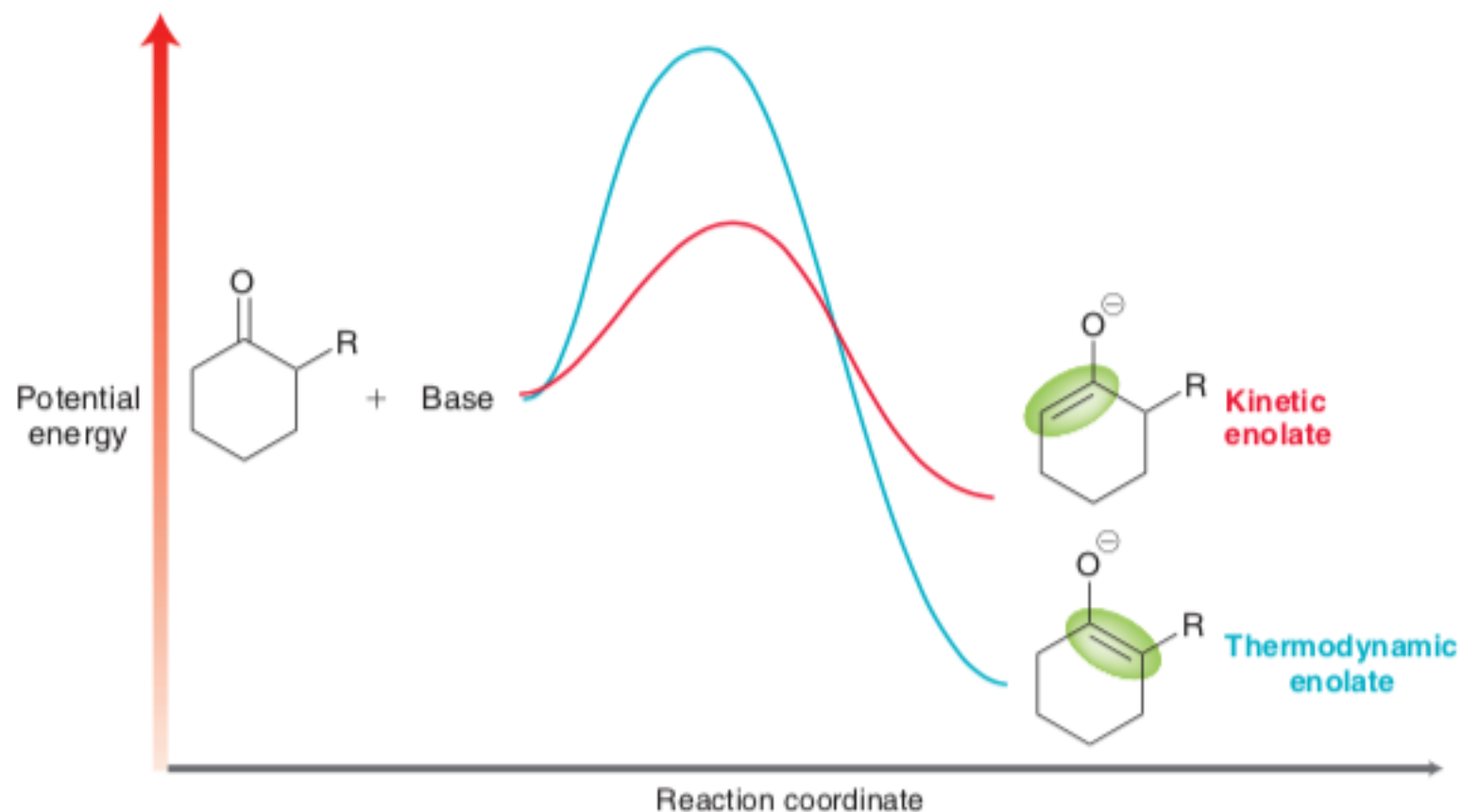
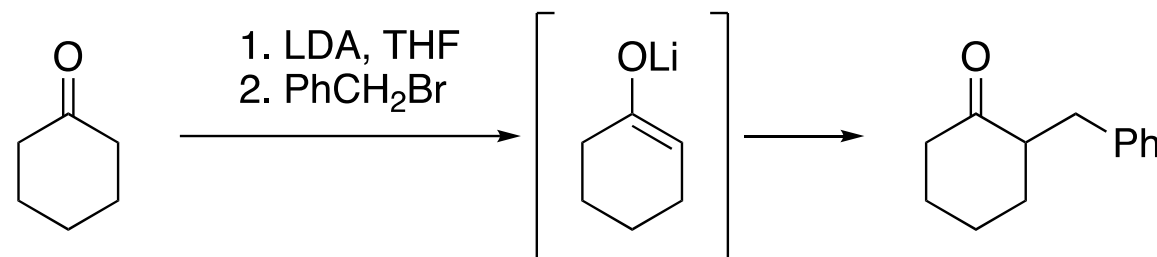


- Basic halogenation of methyl ketones yields carboxylic acids via the **haloform reaction**

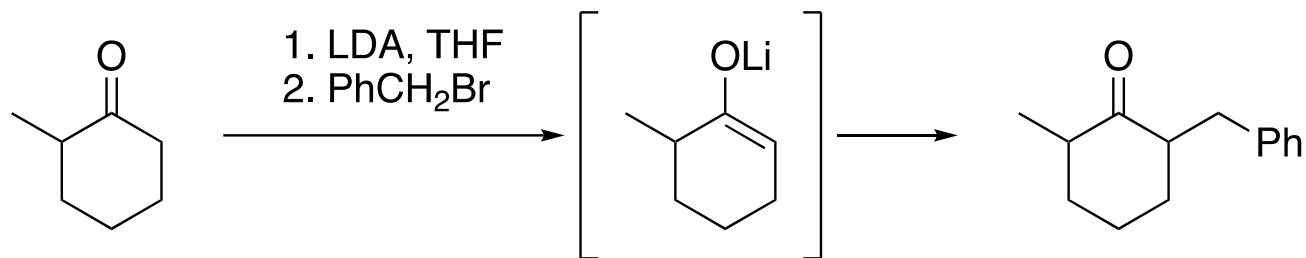


21.5 Alkylation of the Alpha Position | 1 of 2

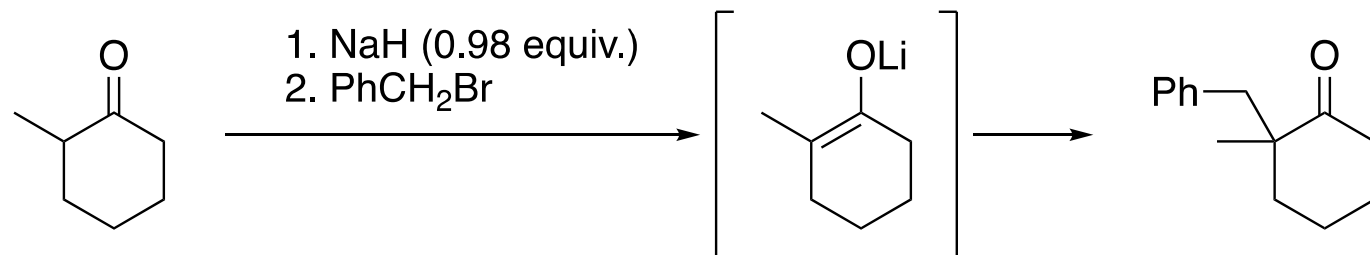
- Enolates can be employed as nucleophiles in S_N2 reactions (*only* methyl or 1° alkyl halides!)
- What if two isomeric enolates are possible?



- Use LDA to quantitatively generate the less substituted enolate (**kinetic enolate**) and then add the alkyl halide



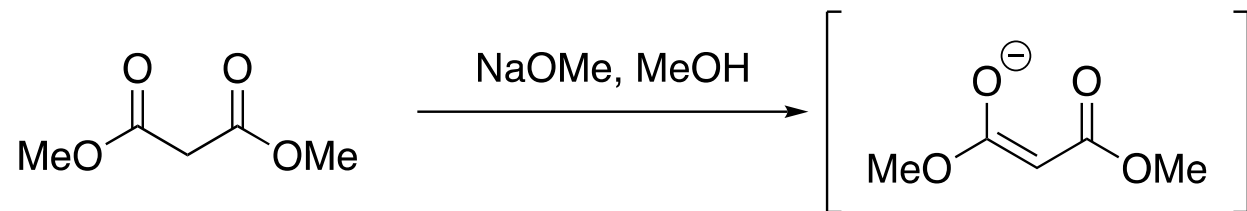
- Use sodium hydride (NaH) at room temperature in slightly less than a full equivalent to generate the more substituted enolate (**thermodynamic enolate**) and then add the alkyl halide



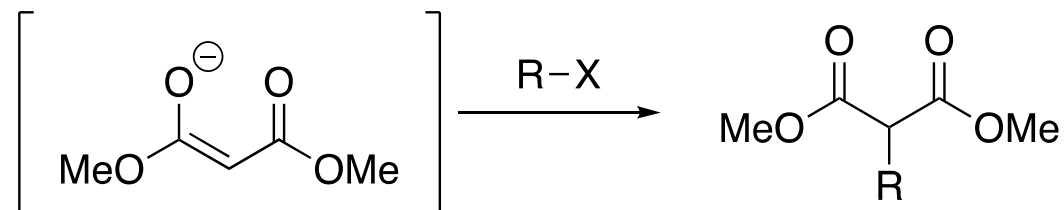
- Or, use the acetoacetic ester synthesis (coming very soon)

- The **malonic ester synthesis** results in the net alkylation of a carboxylic acid

- Malonate esters are easily deprotonated with alkoxide salts (match the alkoxide to the alkoxy groups of the ester!)

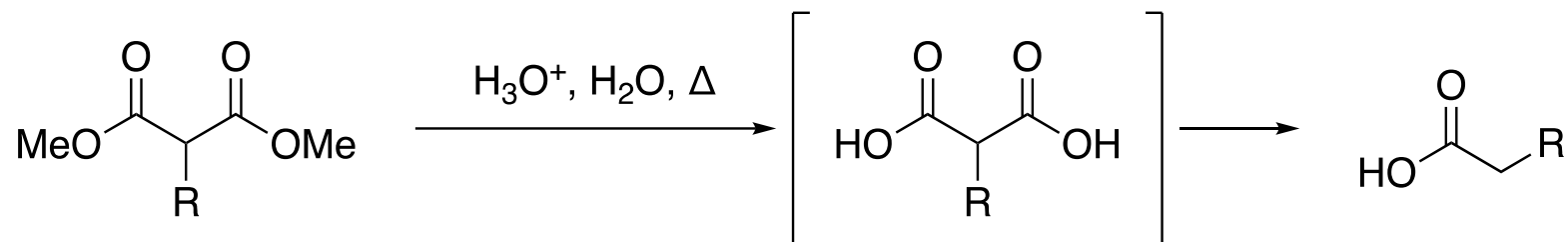


- Stabilized enolate reacts with R-X in an S_N2 reaction

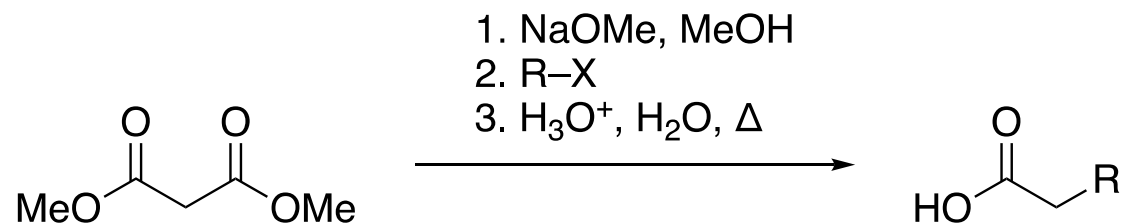


- The first two steps can be repeated to add a second R group, if desired

- The **malonic ester synthesis** results in the net alkylation of a carboxylic acid
4. Treatment with acidic water at high temperatures results in ester hydrolysis and **decarboxylation**

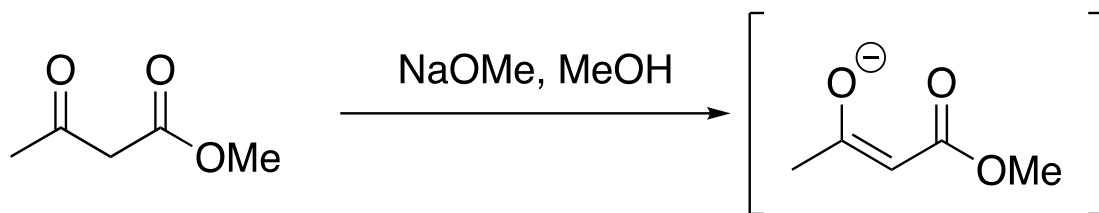


- Overall, the malonate ester is treated with base, alkyl halide, and hot aqueous acid in a three-step sequence; the product is a substituted carboxylic acid

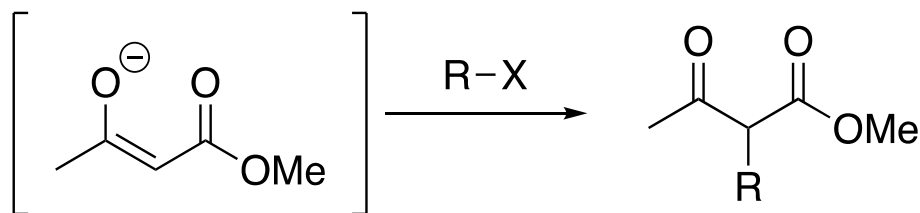


21.5 The Acetoacetic Ester Synthesis | 1 of 2

- The **acetoacetic ester synthesis** results in the net alkylation of acetone
1. Acetoacetic esters are easily deprotonated with alkoxides (match the alkoxide to the alkoxy groups of the ester!)

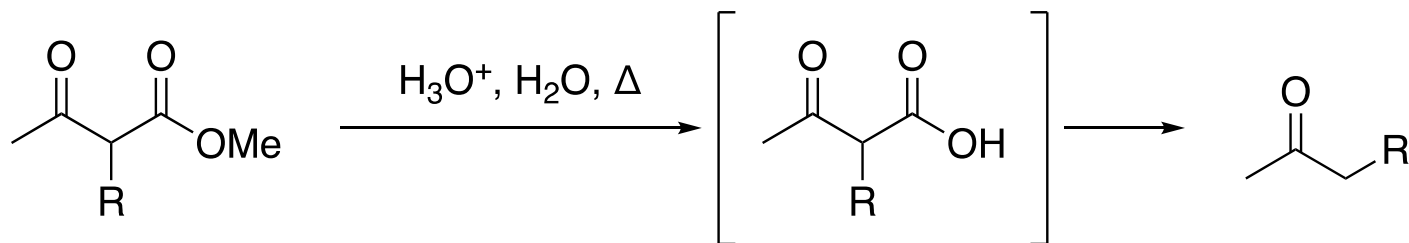


2. Stabilized enolate reacts with R-X in an S_N2 reaction

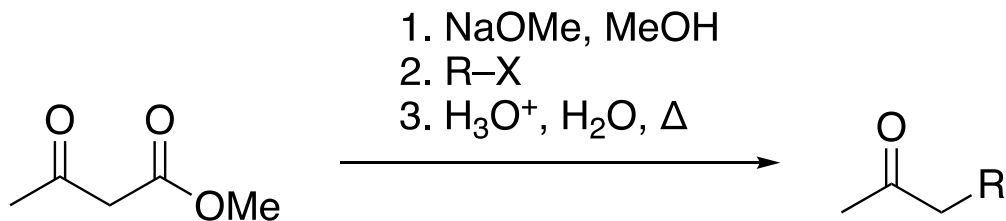


3. The first two steps can be repeated to add a second R group, if desired

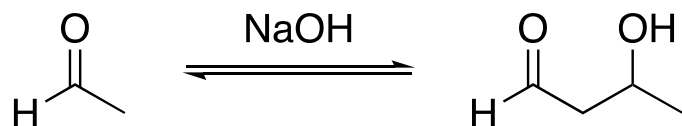
- The **acetoacetic ester synthesis** results in the net alkylation of acetone
4. Treatment with acidic water at high temperatures results in ester hydrolysis and **decarboxylation**



- Overall, the acetoacetic ester is treated with base, alkyl halide, and hot aqueous acid in a three-step sequence; the product is a substituted methyl ketone

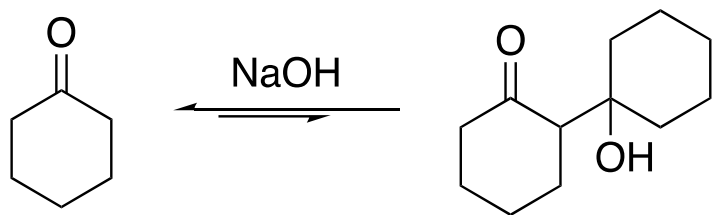


- When an aldehyde is treated with an alkoxide or hydroxide base, a small amount of nucleophilic enolate is in equilibrium with the electrophilic aldehyde
- Addition of the enolate to the aldehyde is called **aldol addition**

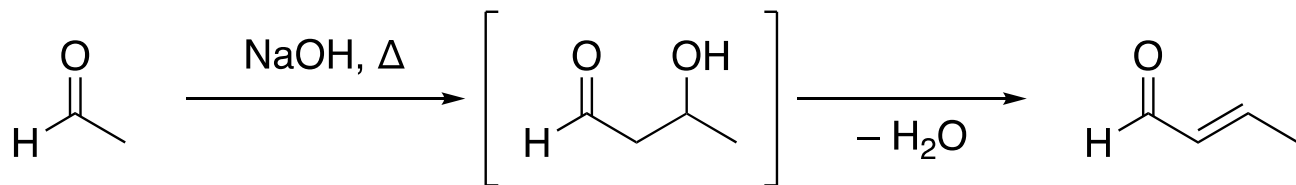


- Reversible deprotonation of the aldehyde (pt) is followed by nucleophilic addition (Ad_N) and protonation of the resulting alkoxide intermediate by water (pt)
- Reaction can also be envisioned for ketones

- The aldol product is thermodynamically favored for aldehydes but disfavored for most ketones for steric reasons

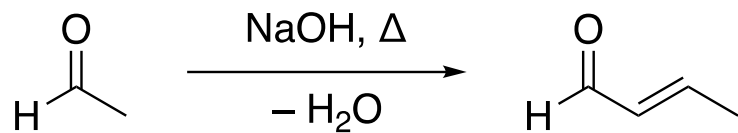


- Fragmentation of a β -hydroxy ketohyde into two carbonyl compounds is called the **retro-aldol reaction**
- When heated under acidic or basic conditions, the β -hydroxy ketohyde will undergo dehydration to yield an α,β -unsaturated ketohyde (**enone** or **enal**); the overall reaction is called **aldol condensation**



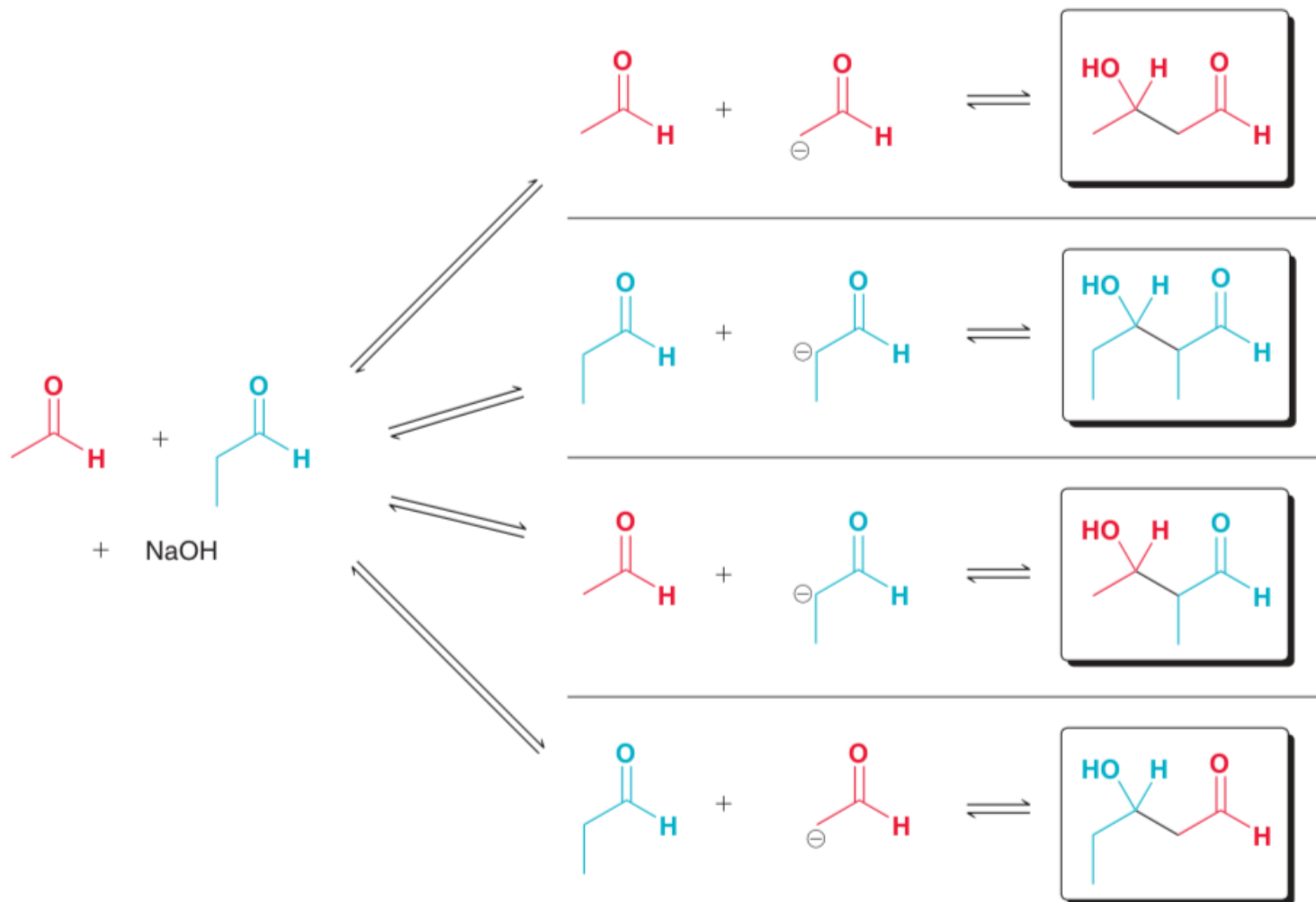
- Driven by the formation of a conjugated system
 - Impossible to stop if condensation would establish an extended conjugated system such as a cinnamyl group
 - Can drive aldol additions of ketones to high yield

- The mechanism of aldol condensation does not involve E2 or E1
- Deprotonation at the α position (pt) is followed by beta-elimination of hydroxide (E_β); the two-step mechanism is collectively called **E1cb**



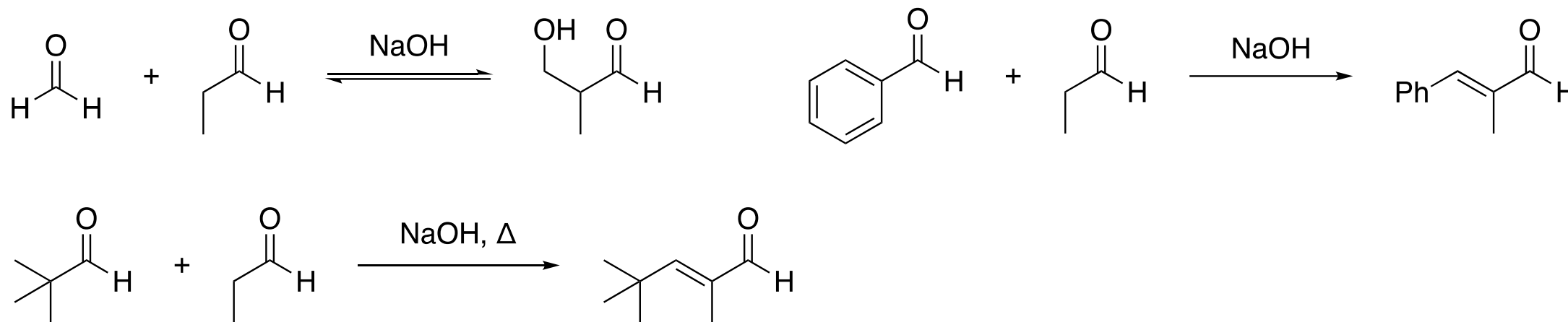
21.3 The Problem with Crossed Aldol Reactions

- Aldol reactions between two different ketohydes can give as many as four products
- Selective formation of a single enolate is not possible—the result is a mess!

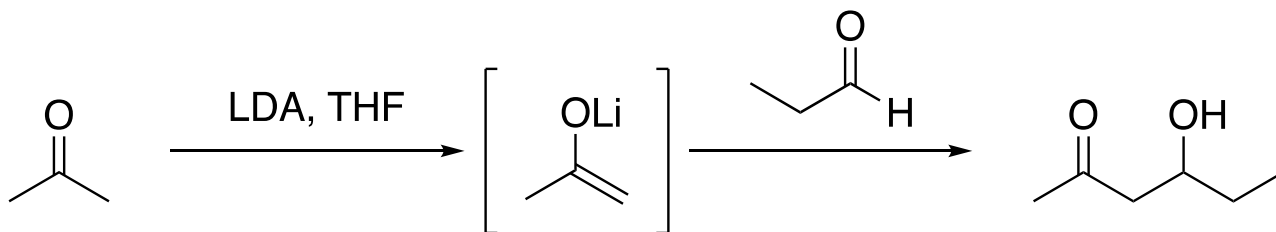


21.3 Claisen-Schmidt Condensation and Directed Aldol Reactions

- One strategy to get around this issue is to use a **non-enolizable** electrophile (**Claisen-Schmidt condensation**)

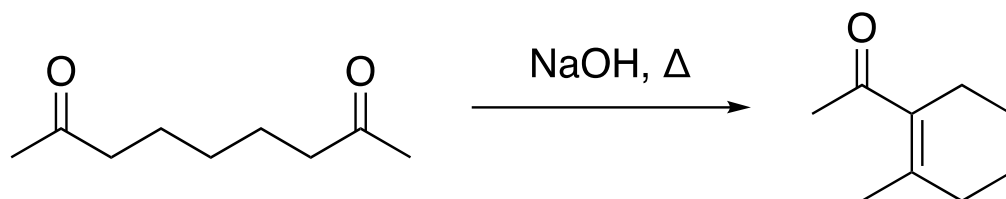
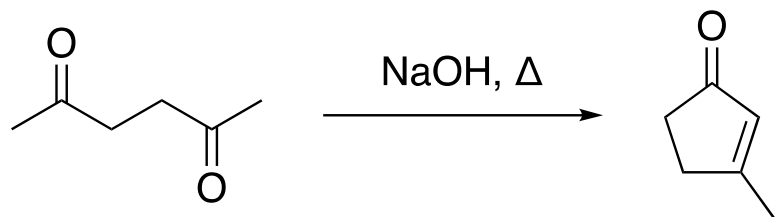


- Complete formation of the enolate with LDA followed by slow addition of the aldehyde electrophile (**directed aldol reaction**) also leads to selective crossed reaction



21.3 Intramolecular Aldol Reactions

- Substrates containing two carbonyl groups can react in **intramolecular aldol reactions**, particularly if the ring formed contains five or six atoms (stable ring sizes)

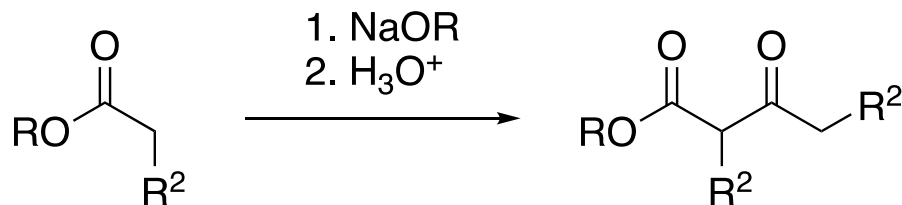


21.4 Acylation of Enolates: The Claisen Condensation

- We have seen enolates act as nucleophiles in S_N2 and nucleophilic addition reactions...can they act as nucleophiles in nucleophilic acyl substitution reactions?



- If we want C-attack (and we do), esters are the electrophile of choice
- Condensation of an ester with a ketone/aldehyde or ester enolate is called the **Claisen condensation**



- Two α -hydrogens in the nucleophile are essential
- Make sure the alkoxide base matches the ester, as in the malonic and acetoacetic ester syntheses!

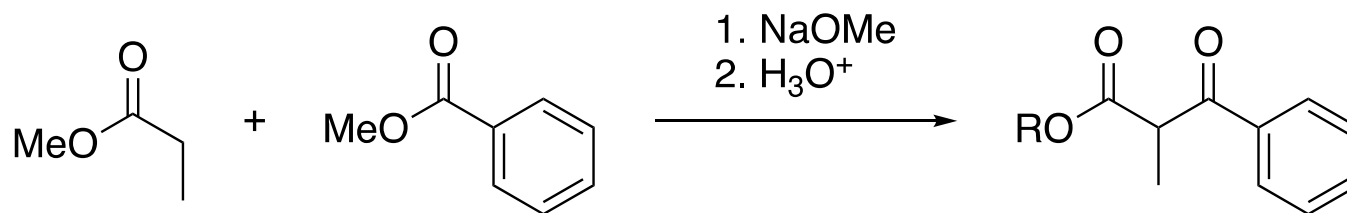
21.4 Mechanism of the Claisen Condensation



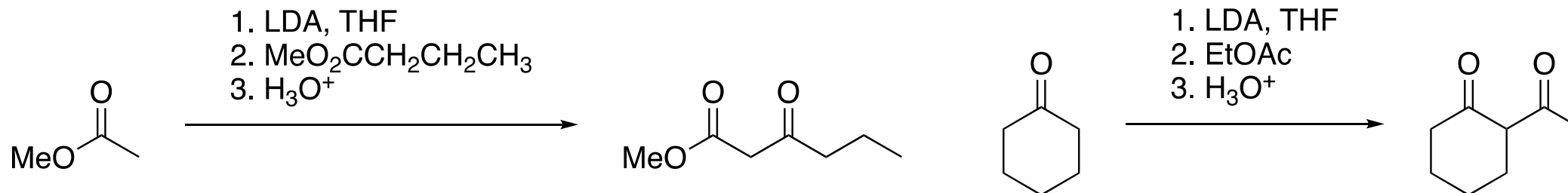
- The mechanism involves enolate formation (pt), nucleophilic addition (Ad_{N}), and beta-elimination (E_{β}); these steps are *thermodynamically unfavorable!*
- The Claisen condensation is rendered favorable by one last proton transfer from the 1,3-dicarbonyl product ($\text{p}K_{\text{a}} < 10$) to the alkoxide base
- Added acid protonates the stabilized enolate, enabling isolation of the neutral product

21.4 Crossed Claisen Condensations

- As in aldol reactions, two different esters generally do not yield a single product when combined in the presence of alkoxide base; instead, the result is a mess
- Two strategies for crossed Claisen condensations:
 - Use of a non-enolizable ester such as a formate, benzoate, or pivalate

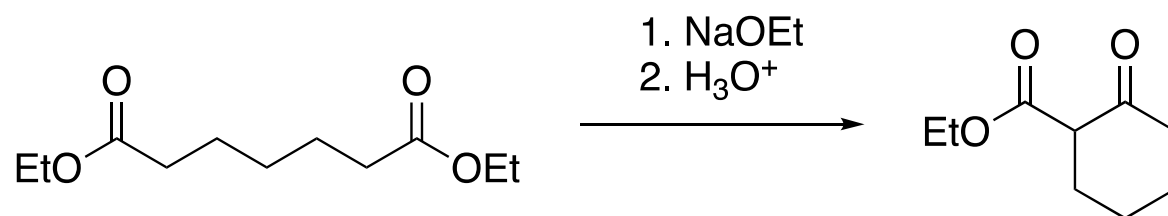


- Directed Claisen condensation, including pre-formation of ketone enolates

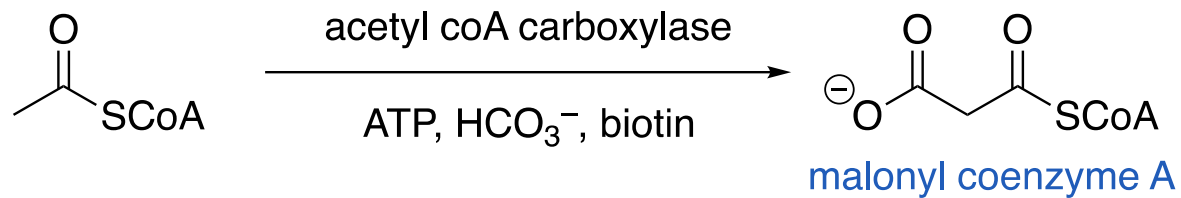


21.4 Intramolecular Claisen (Dieckmann) Condensations

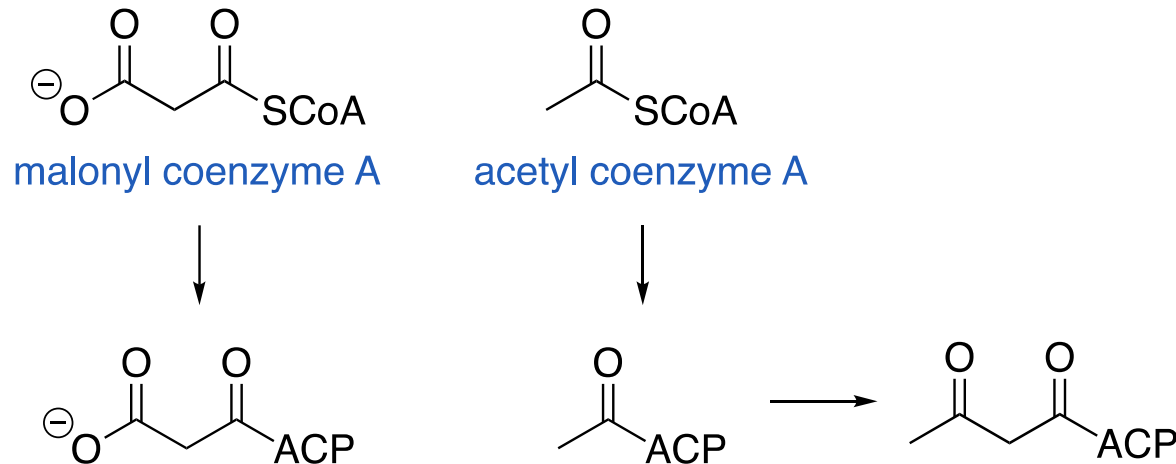
- Intramolecular Claisen condensations are called **Dieckmann condensations** and show a preference for five- and six-membered products, just as in the aldol reaction
- Two α hydrogens on the nucleophilic side are still required



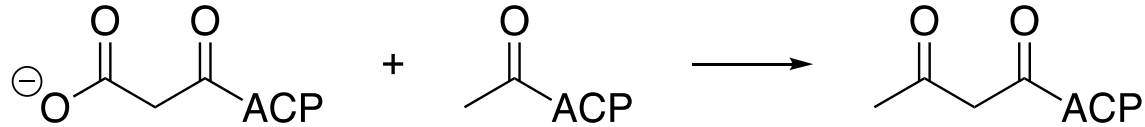
- **Fatty acids** are carboxylic acids containing very long hydrocarbon chains
- They are bio-synthesized starting from acetyl coenzyme A using a series of repeated Claisen condensations
- Malonyl coenzyme A is nature's Claisen nucleophile; acetyl coenzyme A is nature's Claisen electrophile



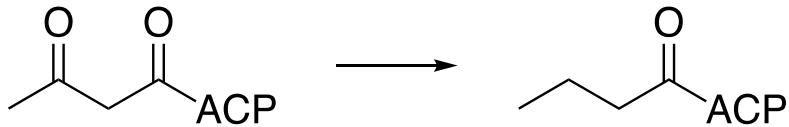
- Both bind to acyl carrier proteins (ACPs) before reacting



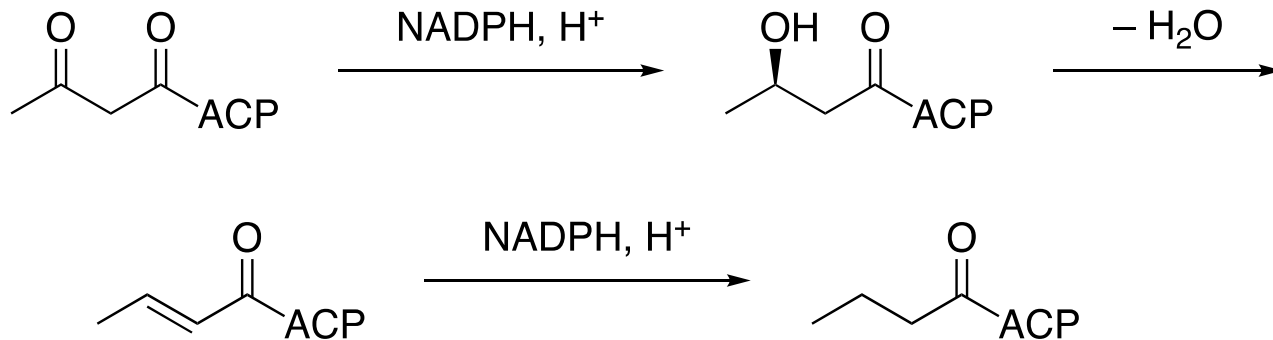
- The mechanism involves decarboxylative formation of an enol from the malonyl ACP followed by nucleophilic acyl substitution



- The ketone group of the β -ketoester must be reduced to $-\text{CH}_2-$ to generate a saturated carbon in the final fatty acid. How would we accomplish this in the laboratory?



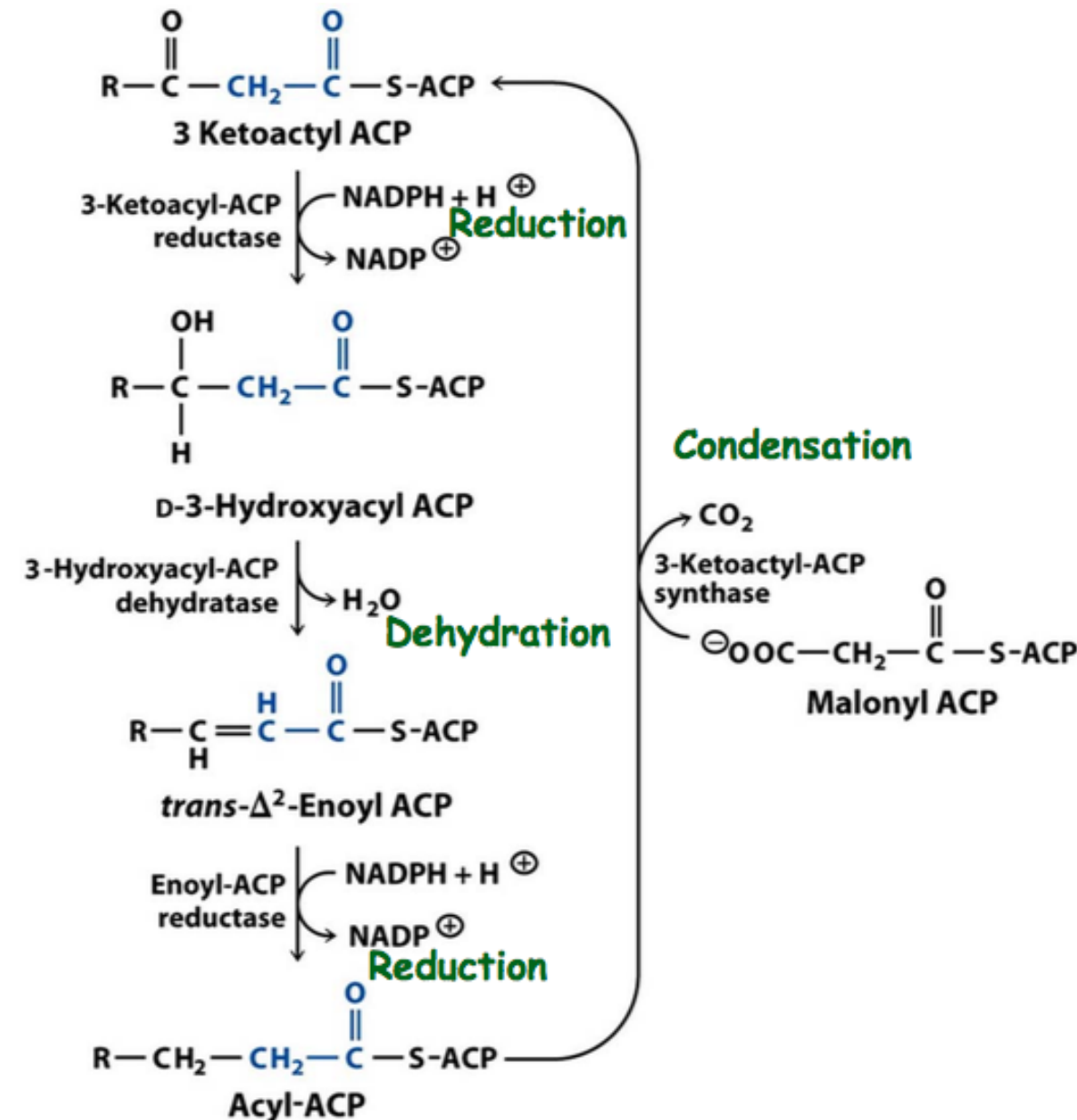
- In the biosynthetic pathway, the β -ketoester is reduced, water is eliminated, and the resulting unsaturated thioester is reduced (hydrogenated)



1. [3-ketoacyl ACP reductase](#)
2. [3-hydroxyacyl ACP dehydratase](#)
3. [enoyl ACP reductase](#)

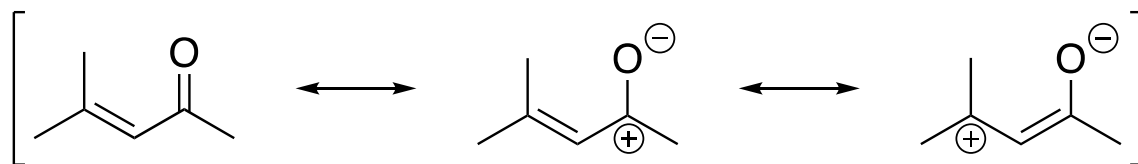
- The resulting elongated acyl ACP is ready to act as an electrophile in the next round; at this point, another molecule of malonyl coenzyme A enters the picture and the cycle restarts

- Repeated cycles elongate the chain, and a final hydrolysis step releases the fatty acid carboxylate
- The structure of this biosynthetic pathway is dictated by fundamental principles of organic chemistry
- For example, why can't acetyl coenzyme A be used directly as the nucleophile? Why is decarboxylation necessary?

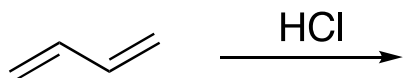


21.6 α,β -Unsaturated Carbonyl Compounds

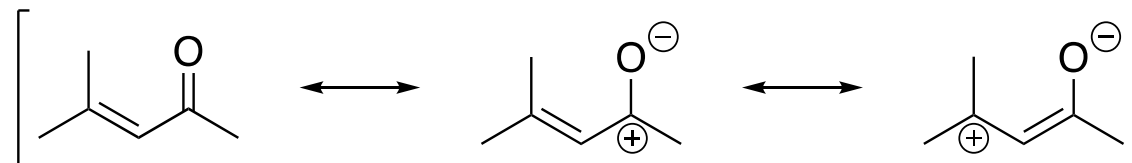
- Carbonyl compounds in which the α and β carbons are linked by a double or triple bond are known as **α,β -unsaturated carbonyl compounds**
- These compounds are electrophilic at *both* the carbonyl carbon *and* the β carbon...resonance shows us why



- Consider an analogy to reactions of conjugated dienes with hydrohalic acids HX



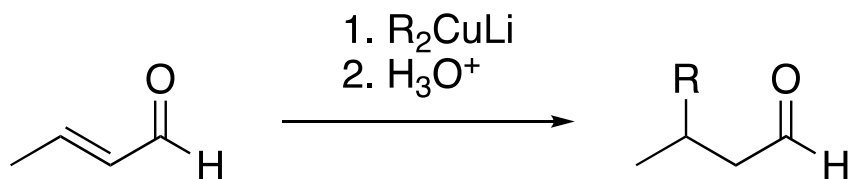
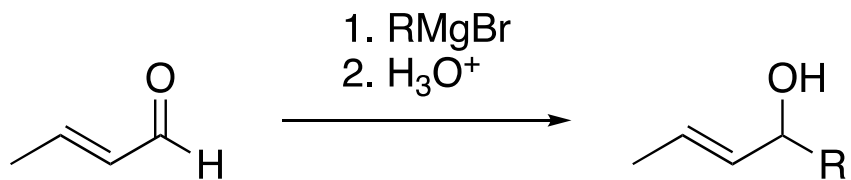
21.6 Nucleophilic Additions to α,β -Unsaturated Carbonyl Compounds



- α,β -Unsaturated carbonyl compounds may engage in two types of nucleophilic addition reactions:
- 1,2-Addition or direct addition
- 1,4-Addition or **conjugate addition**

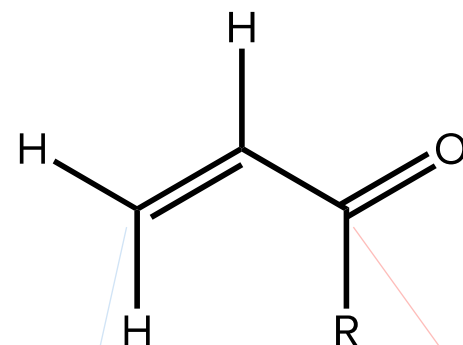
21.6 Direct versus Conjugate Addition

- Three factors affect the relative amounts of 1,2- and 1,4-addition when a nucleophile and unsaturated carbonyl compound are combined
 - How hard or soft is the nucleophile?
 - What is the reaction temperature?
 - How Lewis acidic is the carbonyl carbon?

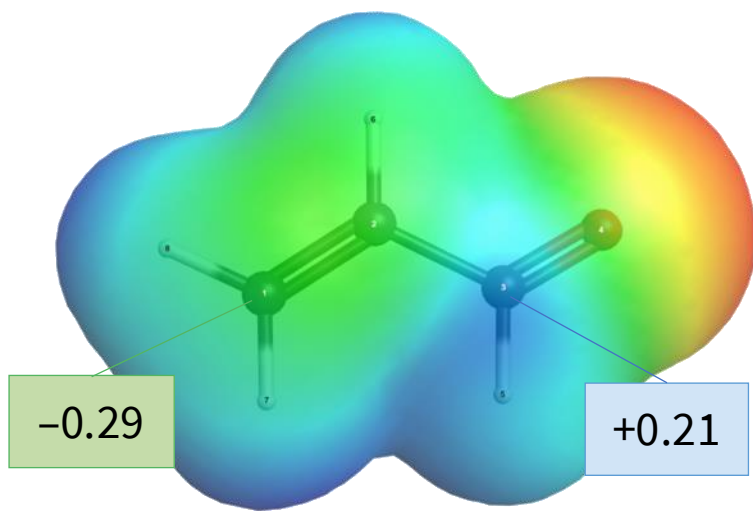


21.6 Hardness/Softness of the Nucleophile

- Small nucleophiles with high charge density (**hard**) add 1,2; large polarizable nucleophiles (**soft**) add 1,4



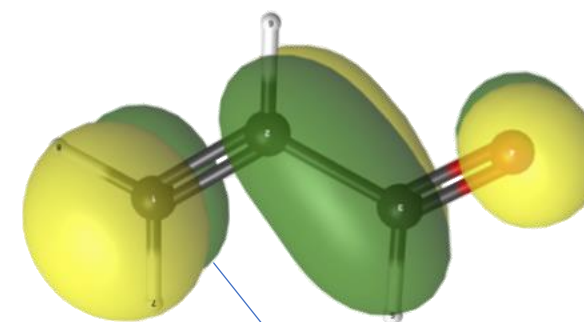
Charge control. Hard nucleophiles are attracted to the partial positive charge of the carbonyl carbon.



Hard nucleophiles
RLi, RMgBr, OH⁻, RO⁻,
ROH, H₂O

Soft nucleophiles
RSH, RS⁻, alkenes, enol(ate)s,
enamines, RCu, R₂CuLi, arenes

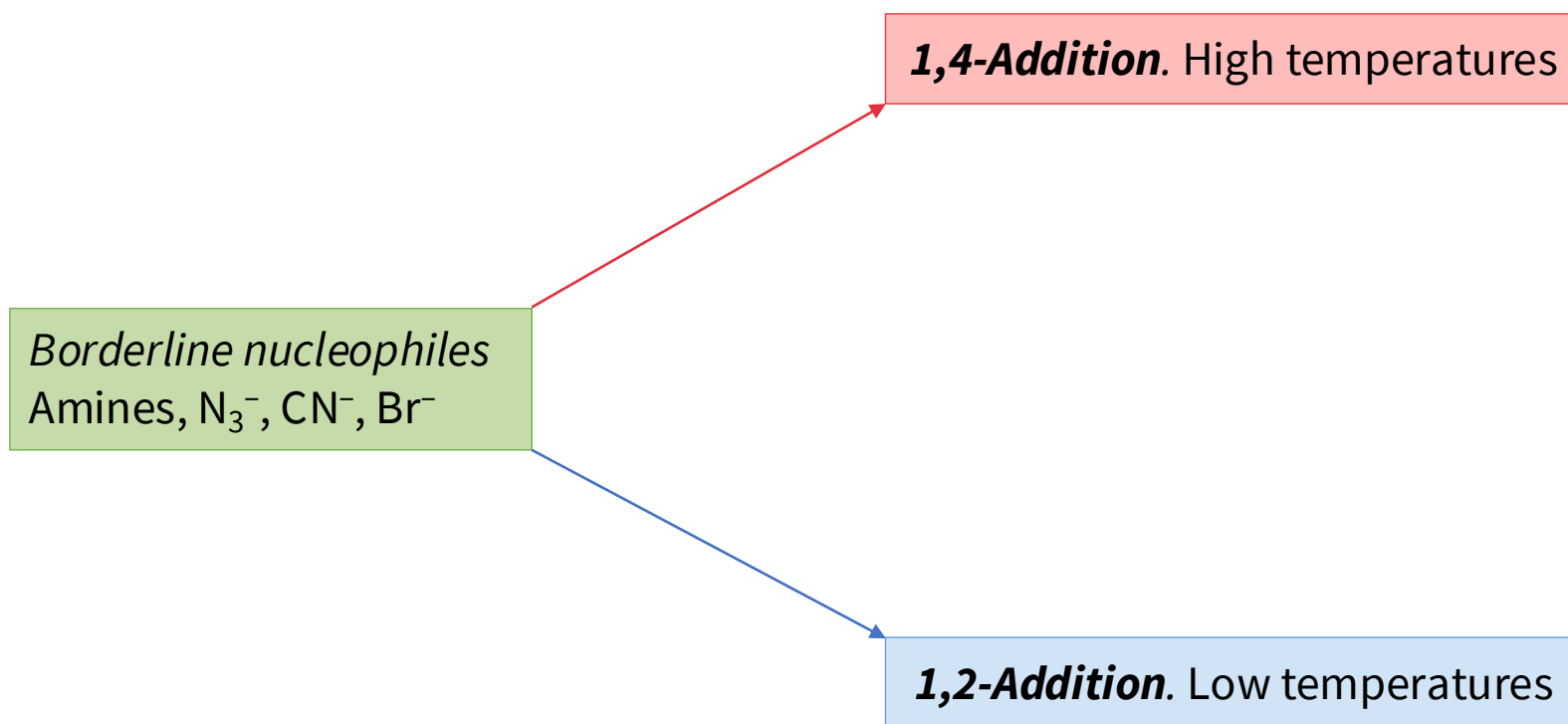
Orbital control. Soft nucleophiles prefer to overlap with the larger lobe in the LUMO at the β -carbon.



Large LUMO density

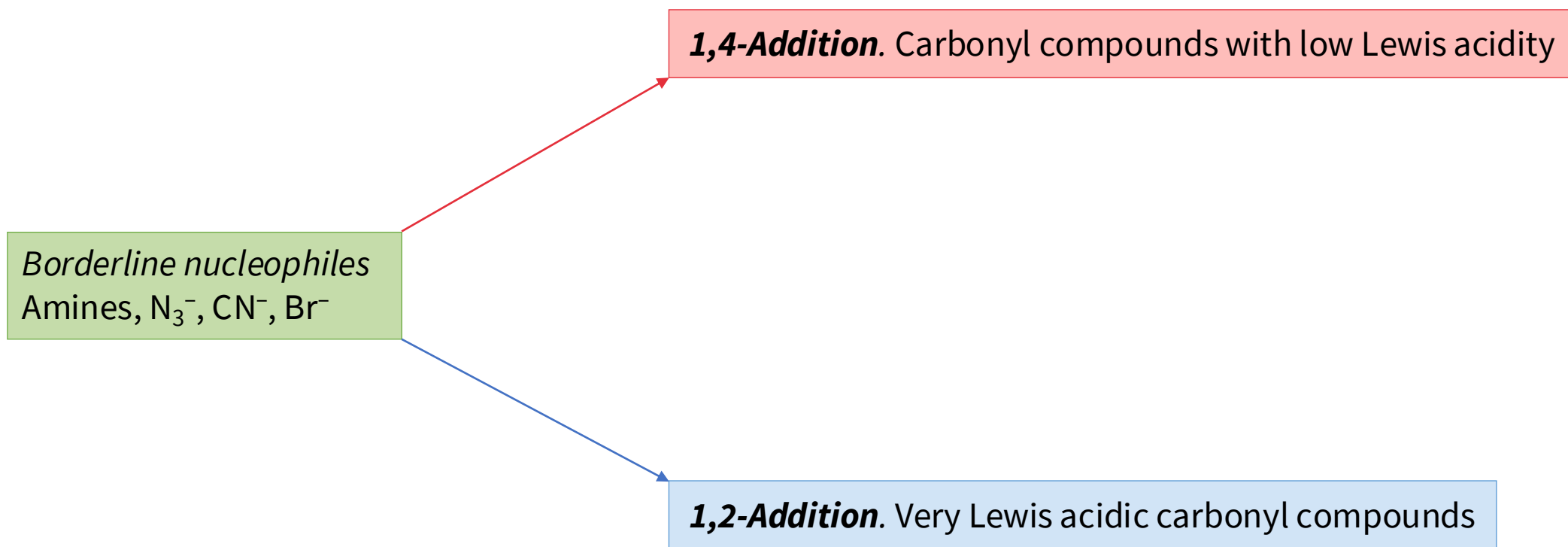
21.6 The Effect of Temperature

- For “borderline” nucleophiles that are not obviously hard or soft and nucleophiles that add reversibly, whether 1,2- or 1,4-addition occurs depends on temperature and the unsaturated carbonyl compound



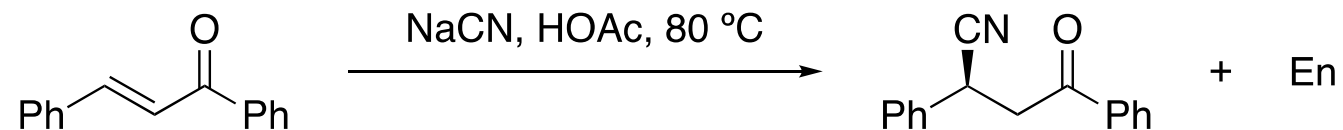
21.6 The Effect of the Unsaturated Carbonyl Compound

- For “borderline” nucleophiles that are not obviously hard or soft and nucleophiles that add reversibly, whether 1,2- or 1,4-addition occurs depends on temperature and the unsaturated carbonyl compound

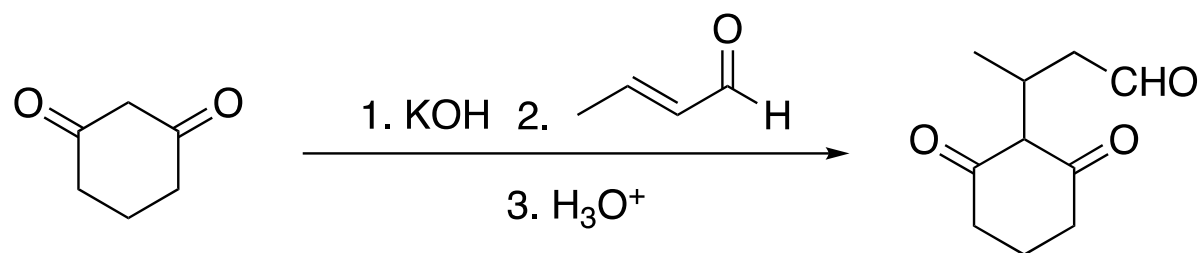


21.6 Conjugate Addition Reactions

- Addition of the nucleophile to the β position yields an enol or enolate intermediate, which tautomerizes rapidly to yield a β -substituted product



- **Stabilized** enolates are soft nucleophiles that add in a conjugate manner to unsaturated carbonyl compounds
- Conjugate addition of an enolate to an unsaturated carbonyl compound is called the **Michael reaction**

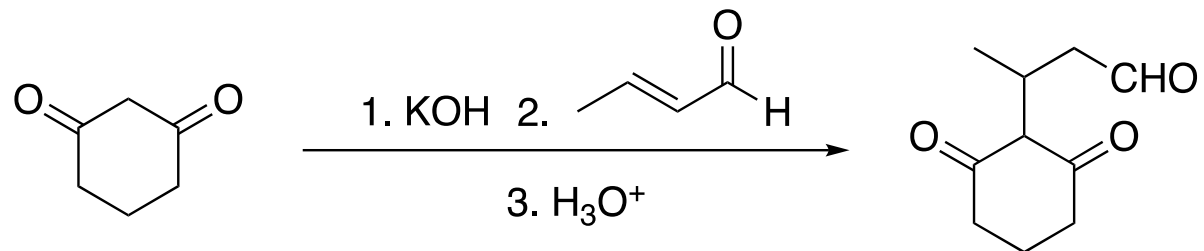


- Stabilized enolates and related stabilized anions are known as **Michael donors**
- Unsaturated carbonyl compounds and related electron-deficient alkenes are known as **Michael acceptors**

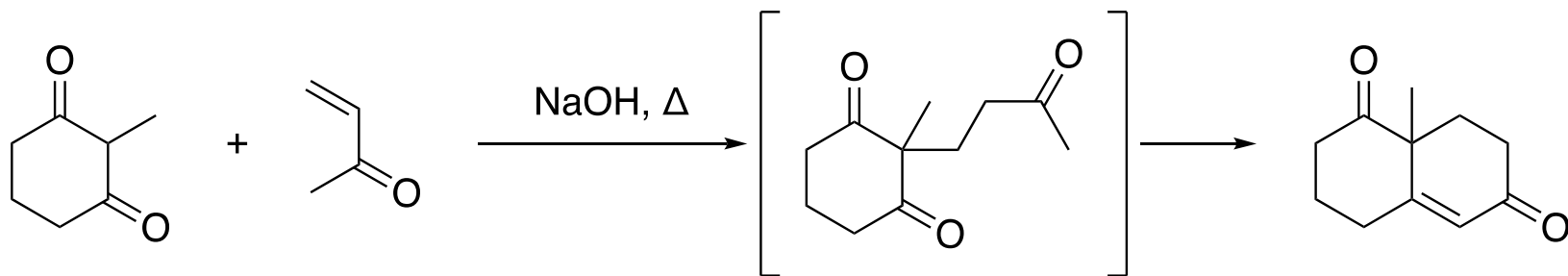
TABLE 21.2 A LIST OF COMMON MICHAEL DONORS AND MICHAEL ACCEPTORS

MICHAEL DONORS		MICHAEL ACCEPTORS	

- The mechanism involves enolate formation (pt) followed by nucleophilic addition (Ad_N) and proton transfer (pt)



- Michael addition generates an enolate...
- If the enolate can add back to the original Michael donor in an aldol condensation, the result is formation of a ring in the product (**Robinson annulation**)
- One C–C bond comes from the Michael addition and the other from the aldol condensation



- To apply the Robinson annulation in retrosynthesis, think about the retro-aldol condensation first and then consider the retro-Michael reaction (the reverse order of the mechanism!)
- Identify reagents that can be used to prepare the following compound via a Robinson annulation:

